Q.29 A circular shaft of 60mm diameter is running at 150rpm. If the shear stress is not to exceed 50MPa, find the power which can be transmitted by the shaft.

Q.30 Define torque and torsion.

Q.31 Draw stress strain curve for a ductile material.

Q.32 Derive an expression for strain energy stored in a material due to suddenly applied load.

Q.33 What is leaf spring? Give its types.

Q.34 Explain temperature stresses.

Q.35 Explain different end conditions for a column.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 A cantilever beam 10m long carries loads of 10Kn, 15 KN and 20 KN at distances of 2m, 5m and 10m respectively from the fixed end. Draw the S.F.D. and B.M.D. for the cantilever.

Q.37 Derive the bending equation by mentioning assumptions taken.

Q.38 Define column. Give its classification and explain factors affecting strength of a column.

Roll No. ....

1st Sem / Mechanical Engineering (MSIL)
Sub : Strength of Materials

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The ability of a material to deform without breaking is called

a) Hardness b) Elasticity

c) Plasticity d) None of the above

Q.2 The unit of strain is

a) KN/m^2 b) m^3

c) No unit d) KNm

Q.3 The polar moment of inertia of a circular section is about

a) Z-Z axis b) X-X axis

c) Y-Y axis d) None of the above

Q.4 In a simply supported beam loaded with point load at centre, the BMD will be

a) A cubic curve b) A triangle

c) A parabolic curve d) None of the above

Q.5 Neutral axis of the beam is the axis.

a) Of maximum stress b) Of zero stress

c) Of negative stress d) All of the above

Q.6 Bending stresses are due to

a) Shear force b) Bending moments

c) Thrust d) None of the above

Q.7 A column whose slenderness ratio is less than 32 is known as

a) Long column b) Short column

c) Medium Colum d) Composite column

Q.8 Euler's formula is generally used when slenderness ratio lies

a) Below 32 b) Above 80

c) Below 80 d) Any value

Q.9 The unit of torque is SI system is

a) N/m^2 b) cm/m^3

c) Nm d) No unit

Q.10 The spring balance spring used is

a) To apply force

b) To absorb shocks

c) To store strain energy

d) To measure force

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define moment of inertia.

Q.12 What do you mean by point of contra flexure?

Q.13 Define lateral strain.

Q.14 Define strain energy.

Q.15 Name any two types of beams.

Q.16 Define concentrated load.

Q.17 Define equivalent length of a column.

Q.18 Write an application of leaf spring.

Q.19 Define bending stress.

Q.20 Define torsion.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain theorem of perpendicular axis.

Q.22 Define elasticity and elastic limit of a material.

Q.23 Explain various types of end supports of beam.

Q.24 Explain strain with its types.

Q.25 Define Poisson's ratio.

Q.26 Write formula for the MOI of a circle and rectangle about its X-X axis.

Q.27 Explain types of loading.

Q.28 Explain Rankine's formula for calculating buckling load.